

ABSTRACT

This project presents a closed-loop obstacle detection and avoidance system for a small autonomous vehicle. A Raspberry Pi 5 camera pipeline converts RGB frames to grayscale, enhances contrast using histogram equalization, and binarizes the image to detect the QR-code signature attached to the obstacle. The detected QR width is used for distance estimation, while EKF-based localization and a Stanley controller generate the steering command required for lane switching. Experiments demonstrated reliable detection up to 60 cm and an average detection time of 0.342 s, allowing the vehicle to switch lanes and avoid the obstacle in repeated trials.

TEAM & PROJECT CONTEXT

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Emails

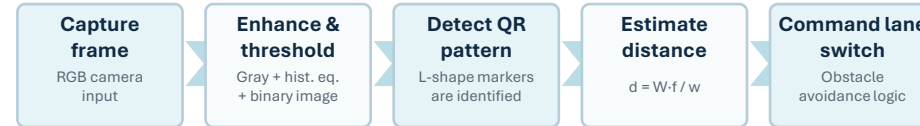
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Objective

Detect a QR-coded obstacle and switch lane autonomously when the object becomes close.

METHODOLOGY

Vision & control pipeline



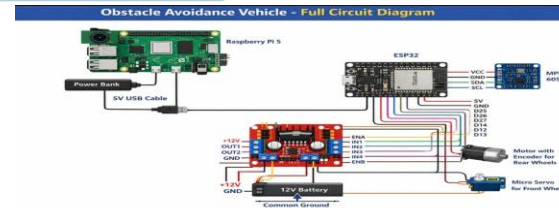
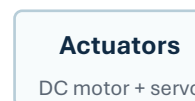
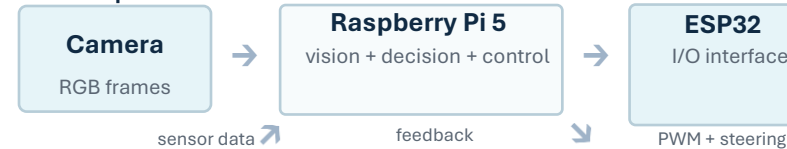
Hardware used in the prototype

- Raspberry Pi 5 + Camera Module 2
- ESP32, IMU-6050, encoder, L298 H-bridge
- DC gear motor, micro-servo, 3D-printed chassis



SYSTEM INTEGRATION

Closed-loop architecture



Distance model

$$d = \frac{W \cdot f}{w}$$

RESULTS & KPI SNAPSHOT



Close obstacle detected
Far obstacle detected
Frames confirm QR localization, distance estimation, and close/far classification.

60 cm

Max detection range

50 cm

Close-obstacle threshold

0.342 s

Average detection time

130 cm

Initial scenario distance

- Repeated trials triggered the correct lane-switch maneuver after obstacle detection.
- Bounding-box visualization confirmed successful QR localization before vehicle reaction.

CONCLUSIONS & LIMITATIONS

- Real-time vision, localization, and control were integrated successfully on the vehicle prototype.
- Limitations: dependence on a QR marker and limited range; future work can generalize detection using richer vision or learning-based models.

COURSE INSTRUCTORS

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